

reopen_20260925

BOB-129 reopen — 2026-09-25

Item: BOB-129 — permanent slowapi rate-limit response: Response header-injection contract regression guard (tests/unit/test_bob129_slowapi_response_contract.py) **Type:** Bug · **Reopen reason:** regression re-discovered by a full bulk pytest run (5033 tests, random seed 12345) plus isolated re-run confirmation, order-independent, reproduced every time. **Fixed again:** 2026-09-25 · **Verified by:** independent subagent re-verification of the conductor's git-history root cause, before applying the fix (§11.4.6 / §11.4.102).

The regression

download-proxy/src/api/routes.py's stream_theme endpoint (GET / theme/stream) was decorated with `@_rl("sse_stream")` in commit 1ef4246 (BOB-167), which added the SSE route to slowapi's rate-limit registry without adding it to BOB-129's hand-curated `_RESPONSE_RETURNING_EXEMPTIONS` allowlist in tests/unit/test_bob129_slowapi_response_contract.py.

BOB-129's contract test

(TestDecoratedEndpointContract::test_every_decorated_endpoint_declares_response_or asserts that every slowapi-decorated endpoint either declares a response: Response parameter, or is a reviewed exemption proven to have exactly one return, always yielding a Response subclass, with every other exit path raising HTTPException (which propagates before slowapi's headers_enabled=True header-injection code ever runs). stream_theme satisfies that shape but was never added to the exemption set, so the test failed.

Independent root-cause re-verification (before applying any fix)

I independently re-read `stream_theme`'s full body in `download-proxy/src/api/routes.py` (lines 162–260) rather than trusting the conductor's claim on its own, per this project's §11.4.6/§11.4.102 investigate-before-fix mandate.

Confirmed by direct inspection + a structural grep over the function's line range:

```
$ awk 'NR==162,NR==260 {print NR": "$0}' download-proxy/src/api/routes.py | grep -nE "return|raise|HTTPException"
50:211:         re-raised by the helper, so structured cancellation
still propagates.
80:241:                 return
89:250:                 # probe-failure return above.
92:253:         return StreamingResponse(
```

Two things had to be checked carefully, not assumed:

1. **The raise hit at line 211 is prose inside a docstring** (“re-raised by the helper”), not executable code — confirmed by reading the surrounding lines. There is no `raise` anywhere in `stream_theme`'s actual body.
2. **The return at line 241 belongs to the nested `async def gen()`: generator function**, not to the outer `stream_theme` route-handler function. Python scopes a `return` to its own enclosing function; `gen()` is a separate function object whose `return` just stops that generator's `async` iteration — it does not touch `stream_theme`'s control flow, and it is not the value `FastAPI/slowapi` wrap as the route's response. The route handler `stream_theme` itself has exactly ONE `return` statement — line 253,
`return StreamingResponse(gen(), media_type=..., headers={...})` — and it is unconditional (no branch, no `raise` precedes or competes with it at the outer-function level).

This is genuinely the same shape as the already-exempted `search_stream` (`download-proxy/src/api/routes.py:1046`), whose body has multiple `raise HTTPException(...)` early-exit paths (404 not found, 403 invalid token, etc.) and exactly one terminal `return SSEHandler.create_streaming_response(...)` — a `StreamingResponse` subclass returned directly, same as `stream_theme`.

Verdict: independent root-cause verification agrees with the conductor's claim in full. No discrepancy found.

Fix applied

Added "api.routes.stream_theme" to _RESPONSE_RETURNING_EXEMPTIONS in tests/unit/test_bob129_slowapi_response_contract.py, with an inline comment (modeled on the existing search_stream comment) stating precisely why the endpoint is exempt.

```
    "api.routes.search_stream",
+   # B0B-129 reopen (2026-09-25): B0B-167 (commit 1ef4246) added
+   #   `@_rl(...)`
+   # to this endpoint without adding it here. `stream_theme`'s
+   #   single
+   # `return` (its ONLY exit path) is
+   #   `StreamingResponse(gen(), ...)`
+   # directly – verified by reading the full function body, no
+   # `raise HTTPException` anywhere in it. The `return` inside
+   #   its nested
+   # `gen()` async generator belongs to that inner generator's
+   #   own scope,
+   # not to `stream_theme` itself, and does not change the outer
+   # function's single-return-path shape. Same exemption
+   #   reasoning as
+   # `search_stream` above.
+   "api.routes.stream_theme",
}
```

No changes were needed in download-proxy/src/api/routes.py — stream_theme's implementation is correct as-is; the fix is entirely in the test file's exemption set.

RED → GREEN → RED → GREEN proof (§11.4.43 / §11.4.115)

Step 1 — RED baseline (before any fix), confirming the regression genuinely reproduces

```
$ .venv/bin/python -m pytest tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
-V
...
>         assert not offenders, (
>             "slowapi-decorated endpoint(s) declare no `response:
Response` parameter and "
```

```

        "are not a reviewed always-returns-Response exemption.
With "
        "`headers_enabled=True` these raise "
        f"{_SLOWAPI_CONTRACT_ERROR!r} on every request whose
handler returns a "
        f"non-Response (BOB-129): {offenders}"
    )
E       AssertionError: slowapi-decorated endpoint(s) declare no
`response: Response` parameter and are not a reviewed always-
returns-Response exemption. With `headers_enabled=True` these
raise 'parameter `response` must be an instance of
starlette.responses.Response' on every request whose handler
returns a non-Response (BOB-129): ["api.routes.stream_theme
(params: ['request'])"]
E       assert not ["api.routes.stream_theme (params:
['request'])"]

tests/unit/test_bob129_slowapi_response_contract.py:392:
AssertionError
===== warnings summary
=====
.venv/lib/python3.12/site-packages/fastapi/testclient.py:1
/home/milosvasic/Projects/boba/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning:
Using `httpx` with `starlette.testclient` is deprecated; install
`httpx2` instead.
    from starlette.testclient import TestClient as TestClient #
noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-
warnings.html
===== short test summary info
=====
FAILED tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
===== 1 failed, 1 warning in 0.69s
=====

```

Step 2 — Fix applied, GREEN confirmed

```

$ .venv/bin/python -m pytest tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
-v
...
tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
PASSED [100%]
...
===== 1 passed, 1 warning in 0.57s
=====

```

Step 3 — Fix temporarily reverted (the "api.routes.stream_theme" line commented

out under a marker), confirming RED again — proving it is genuinely this specific exemption entry that makes the test pass, not something else

```
$ .venv/bin/python -m pytest tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
-v
...
>         assert not offenders, (
...
        )
E         AssertionError: slowapi-decorated endpoint(s) declare no
`response: Response` parameter and are not a reviewed always-
returns-Response exemption. With `headers_enabled=True` these
raise 'parameter `response` must be an instance of
starlette.responses.Response' on every request whose handler
returns a non-Response (BOB-129): ["api.routes.stream_theme
(params: ['request'])"]
E         assert not ["api.routes.stream_theme (params:
['request'])"]

tests/unit/test_bob129_slowapi_response_contract.py:393:
AssertionError
===== warnings summary =====
.venv/lib/python3.12/site-packages/fastapi/testclient.py:1
/home/milosvasic/Projects/boba/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning:
Using `httpx` with `starlette.testclient` is deprecated; install
`httpx2` instead.
    from starlette.testclient import TestClient as TestClient #
noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-
warnings.html
===== short test summary info =====
FAILED tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
===== 1 failed, 1 warning in 0.60s =====
```

Step 4 — Fix restored (exemption entry with its full comment re-added), GREEN

confirmed again — final state

```
$ .venv/bin/python -m pytest tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
-v
...
tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
PASSED [100%]
...
===== 1 passed, 1 warning in 0.57s
=====
```

RED→GREEN→RED→GREEN: the test unambiguously catches the regression (fails when the `stream_theme` exemption is absent) and unambiguously passes once, and only once, that specific exemption entry is present — not a coincidental side effect of anything else.

Full-file regression check — zero other regressions

```
$ .venv/bin/python -m pytest tests/unit/
test_bob129_slowapi_response_contract.py -v
...
collected 7 items

tests/unit/
test_bob129_slowapi_response_contract.py::TestProductionDashboardD:
PASSED [ 14%]
tests/unit/
test_bob129_slowapi_response_contract.py::TestProductionDashboardD:
dashboard] PASSED [ 28%]
tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
PASSED [ 42%]
tests/unit/
test_bob129_slowapi_response_contract.py::TestDecoratedEndpointCont
PASSED [ 57%]
tests/unit/
test_bob129_slowapi_response_contract.py::TestGuardCannotSilentlySl
PASSED [ 71%]
tests/unit/
test_bob129_slowapi_response_contract.py::TestControlNeedle::test_u
PASSED [ 85%]
tests/unit/
```

```
test_bob129_slowapi_response_contract.py::TestControlNeedle::test_c
PASSED [100%]
```

```
===== warnings summary
```

```
=====
```

```
.venv/lib/python3.12/site-packages/fastapi/testclient.py:1
/home/milosvasic/Projects/boba/.venv/lib/python3.12/site-
packages/fastapi/testclient.py:1: StarletteDeprecationWarning:
Using `httpx` with `starlette.testclient` is deprecated; install
`httpx2` instead.
    from starlette.testclient import TestClient as TestClient #
noqa
```

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
```

```
===== 7 passed, 1 warning in 1.36s
=====
```

7/7 passed, 0 failed, 0 skipped.

Files touched

- tests/unit/test_bob129_slowapi_response_contract.py — added "api.routes.stream_theme" to _RESPONSE_RETURNING_EXEMPTIONS with an inline justification comment.
- download-proxy/src/api/routes.py — **not modified**. stream_theme's implementation is correct; no source change was required.
- docs/qa/B0B-129/reopen_20260925.md — this evidence file (new).

Out of scope, not touched by this fix (per task constraints): docs/workable_items.db, docs/Issues.md, docs/Fixed.md, any workable-items CLI command, scripts/pre_build_verification.sh, scripts/commit-push-all.sh. Tracker/DB reconciliation for this reopen is the conductor's responsibility per this project's §11.4.214 recurrence-links-not-mints rule.